

Identifying Existing Users

When we migrate the database, Django will modify the database so it can store a connection between each topic and a user. To make the migration, Django needs to know which user to associate with each existing topic. The simplest approach is to start by assigning all existing topics to one user—for example, the superuser. But first, we need to know that user’s ID.

Let’s look at the IDs of all users created so far. Start a Django shell session and issue the following commands:

```
(ll_env)learning_log$ python manage.py shell
❶ >>> from django.contrib.auth.models import User
❷ >>> User.objects.all()
<QuerySet [<User: ll_admin>, <User: eric>, <User: willie>]>
❸ >>> for user in User.objects.all():
...     print(user.username, user.id)
...
ll_admin 1
eric 2
willie 3
>>>
```

We first import the `User` model into the shell session ❶. We then look at all the users that have been created so far ❷. The output shows three users for my version of the project: `ll_admin`, `eric`, and `willie`.

Next, we loop through the list of users and print each user’s username and ID ❸. When Django asks which user to associate the existing topics with, we’ll use one of these ID values.

Migrating the Database

Now that we know the IDs, we can migrate the database. When we do this, Python will ask us to connect the `Topic` model to a particular owner temporarily or to add a default to our *models.py* file to tell it what to do. Choose option 1:

```
❶ (ll_env)learning_log$ python manage.py makemigrations learning_logs
❷ It is impossible to add a non-nullable field 'owner' to topic without
specifying a default. This is because...
❸ Please select a fix:
  1) Provide a one-off default now (will be set on all existing rows with a
    null value for this column)
  2) Quit and manually define a default value in models.py.
❹ Select an option: 1
❺ Please enter the default value now, as valid Python
The datetime and django.utils.timezone modules are available...
Type 'exit' to exit this prompt
❻ >>> 1
Migrations for 'learning_logs':
  learning_logs/migrations/0003_topic_owner.py
- Add field owner to topic
(ll_env)learning_log$
```
